```
Using device: cpu
Using existing raw data from /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML
/data_testing/sqed_4pt_physical/sqed_4pt_physical_prepared.csv
Imported 1 / 1 rows
Wrote raw data to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testi
ng/outputs/sqed_4pt_physical_prepared_beam20_eval.csv
Wrote tokenised data to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data
_testing/outputs/sqed_4pt_physical_prepared_beam20_eval_tok.csv
Dedupe removed 0 duplicate pairs
Input token limit: unlimited; longest input: 61 content / 63 with BOS/EOS
Numerical evaluator: sqed (mass=2.0; pol_modes=coulomb,covariant; tol_abs=1e-12; tol_rel=1e-10)
Loading model from /Users/kymani/Documents/SQED_KYMANI/SQED/4PT_Model/best_model.pt

Running decode mode: beam; max_length=256; beam_size=20; mask_invalid_tokens=False
  wrote human CSV to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_te
sting/outputs/sqed_4pt_physical_prepared_beam20_eval_beam_human.csv

[beam]  n=1
  top-1 equivalent   :    1 / 1  (100.0%)
  any-beam equivalent:    1 / 1  (100.0%)
  avg valid beams : 15.0
  shorter equivalent outputs: 1 / 1
  copied inputs: 0 / 1
  avg tokens (correct):  pred 18.0  vs  scrambled 61.0

Example predictions (1 rows):
  row 0 [beam]
    target simple : −2*(p_1 · F_2 · F_3 · p_1)/((p_1 · p_2)*(p_1 · p_3))
    input scram   : 2*(e_2 · e_3) − 2*(e_2 · p_1)*(e_3 · p_1)/(p_1 · p_2) − 2*(e_2 · p_1)*(e_3 · p_2
)/(p_1 · p_2) − 2*(e_3 · p_1)*(e_2 · p_1)/(p_1 · p_3) − 2*(e_3 · p_1)*(e_2 · p_3)/(p_1 · p_3)
    top1 pred     : (−2*p_1·F_2·F_3·p_1)/(p_1·p_2*p_1·p_3)
    top1 metrics  : token=False string=True num_simple=True num_scr=1
    any-beam      : token=0 string=1 num_simple=1 num_scr=1
  wrote detail results to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/da
ta_testing/outputs/sqed_4pt_physical_prepared_beam20_eval_beam_results.csv

Wrote summary to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testin
g/outputs/sqed_4pt_physical_prepared_beam20_eval_summary.csv
Total wall time: 1.69s

Terminal output PDF: /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_te
sting/outputs/SQED_4_Evaluation_Information_2026-10-07_17-12-19-729585.pdf
```